Linear Algebra: Homework 1

Yifan Chen

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Problem 1 [10 pts]

The augmented matrix of a linear system has been reduced by row operations to the form shown below. Continue the appropriate row operations (till its reduced echelon form) and describe the solution set of the system.

$$\begin{bmatrix} 1 & -5 & 4 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \end{bmatrix}$$

Answer:

Let $\boldsymbol{a}_i$ be the i-th row of the matrix. We do the following operations to reduce the matrix:

- $\boldsymbol{a}_4 \leftarrow \frac{1}{2} \cdot \boldsymbol{a}_4$;
- $\boldsymbol{a}_3 \leftarrow \frac{1}{3} \cdot \boldsymbol{a}_3$;
- $\boldsymbol{a}_2 \leftarrow \boldsymbol{a}_2 + (-1) \cdot \boldsymbol{a}_4$;
- $\boldsymbol{a}_1 \leftarrow \boldsymbol{a}_1 + 5 \cdot \boldsymbol{a}_2 + (-4) \cdot \boldsymbol{a}_3$.

Then the matrix becomes

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

Therefore, the solution set of the system is $x_1 = x_2 = x_3 = x_4 = 0$.

Problem 2 [10 pts]

Solve the following system of linear equations:

$$\begin{aligned} x_1 \quad\quad\quad -3x_3 &= 8 \\ 2x_1 + 2x_2 + 9x_3 &= 7 \\ + x_2 + 5x_3 &= -2 \end{aligned}$$

Answer:

We first write the linear equations into matrix form:

$$\begin{bmatrix} 1 & 0 & -3 & 8 \\ 2 & 2 & 9 & 7 \\ 0 & 1 & 5 & -2 \end{bmatrix}.$$

Let $\boldsymbol{a}_i$ be the i-th row of the matrix. We do the following operations to solve the equations:

- $\boldsymbol{a}_2 \leftarrow \boldsymbol{a}_2 - 2 \cdot \boldsymbol{a}_1$, now $\boldsymbol{a}_2 = (0, 2, 15, -9)$;
- $\boldsymbol{a}_3 \leftarrow \boldsymbol{a}_3 - \frac{1}{2} \cdot \boldsymbol{a}_2$, now $\boldsymbol{a}_3 = (0, 0, -\frac{5}{2}, \frac{5}{2})$;
- $\boldsymbol{a}_3 \leftarrow -\frac{2}{5} \cdot \boldsymbol{a}_3$, now $\boldsymbol{a}_3 = (0, 0, 1, -1)$;
- $\boldsymbol{a}_2 \leftarrow \frac{1}{2} \cdot \boldsymbol{a}_2 - \frac{15}{2} \cdot \boldsymbol{a}_3$, now $\boldsymbol{a}_2 = (0, 1, 0, 3)$;
- $\boldsymbol{a}_1 \leftarrow \boldsymbol{a}_1 + 3 \cdot \boldsymbol{a}_3$, now $\boldsymbol{a}_1 = (1, 0, 0, 5)$.

Therefore, the solution to the equations is

$$\begin{cases} x_1 = 5, \\ x_2 = 3, \\ x_3 = -1. \end{cases}$$

Problem 3 [8 pts]

True or False. You don't need to justify your answer.

1. Two matrices are row equivalent if they have the same number of rows.

2. Elementary row operations on an augmented matrix never change the solution set of the associated linear system.

3. Two equivalent linear systems can have different solution sets.

4. A consistent system of linear equations has one or more solutions.

Answer:

The answers to these problems are, in order, F, T, F, T.

Problem 4 [8 pts]

For the following matrices, determine which matrices are in reduced echelon form and which others are only in echelon form.

a. $\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&1\end{bmatrix}$

b. $\begin{bmatrix}1&0&1&0\\0&1&1&0\\0&0&0&1\end{bmatrix}$

c. $\begin{bmatrix}1&0&0&0\\0&1&1&0\\0&0&0&0\\0&0&0&1\end{bmatrix}$

d. $\begin{bmatrix}1&1&0&1&1\\0&2&0&2&2\\0&0&0&3&3\\0&0&0&0&4\end{bmatrix}$

Answer:

The matrices a. and b. are in reduced echelon form, and matrix d. is only in echelon form since there are non-1 leading entries in nonzero rows. Matrix c. is not in echelon form since there is a zero row above the nonzero row.

Problem 5 [10 pts]

Row reduce the matrix to reduced echelon form. Circle the pivot positions in the final matrix and in the original matrix, and list the pivot columns.

$$\begin{bmatrix}1&2&4&8\\2&4&6&8\\3&6&9&12\end{bmatrix}$$

Answer:

Let $\boldsymbol{a}_i$ be the i-th row of the matrix. We do the following operations to reduce the matrix:

- $\boldsymbol{a}_2 \leftarrow \boldsymbol{a}_2 - 2\cdot\boldsymbol{a}_1$, now $\boldsymbol{a}_2 = (0,0,-2,-8)$;
- $\boldsymbol{a}_3 \leftarrow \boldsymbol{a}_3 - 3\cdot\boldsymbol{a}_1 - \frac{3}{2}\cdot\boldsymbol{a}_2$, now $\boldsymbol{a}_3 = (0,0,0,0)$;
- $\boldsymbol{a}_2 \leftarrow -\frac{1}{2}\cdot\boldsymbol{a}_2$, now $\boldsymbol{a}_2 = (0,0,1,4)$;
- $\boldsymbol{a}_1 \leftarrow \boldsymbol{a}_1 - 4\cdot\boldsymbol{a}_2$, now $\boldsymbol{a}_1 = (1,2,0,-8)$.

Then the matrix becomes

$$\begin{bmatrix}\boxed{1}&2&0&-8\\0&0&\boxed{1}&4\\0&0&0&0\end{bmatrix},$$

while the boxed numbers are pivot positions and the first and the third columns are pivot columns.

The pivot positions in the original matrix are

$$\begin{bmatrix}\boxed{1}&2&4&8\\2&4&\boxed{6}&8\\3&6&9&12\end{bmatrix}.$$

Problem 6 [5 pts]

Find the general solutions of the system whose augmented matrix is given below.

$$\begin{bmatrix} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{bmatrix}$$

Answer:

Reducing the matrix by the same method we used in previous problems, we obtain the reduced echelon form:

$$\begin{bmatrix} 1 & 3 & 0 & -5 \\ 0 & 0 & 1 & 3 \end{bmatrix}.$$

Therefore, the solution set of the system is $x_2 \in \mathbb{R}, x_1 = -5 - 3x_2, x_3 = 3$.

Problem 7 [10 pts]

Find the general solutions of the system whose augmented matrix is given below.

$$\begin{bmatrix} 3 & -2 & 4 & 0 \\ 9 & -6 & 12 & 0 \\ 6 & -4 & 8 & 0 \end{bmatrix}$$

Answer:

Reducing the matrix by the same method we used in previous problems, we obtain the reduced echelon form:

$$\begin{bmatrix} 1 & -\frac{2}{3} & \frac{4}{3} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Therefore, the solution set of the system is $x_2, x_3 \in \mathbb{R}, x_1 = \frac{2}{3}x_2 - \frac{4}{3}x_3$.

Problem 8 [8 pts]

The following matrices are in echelon form. The leading entries (■) may have any nonzero value; the starred entries (∗) may have any value (including zero). Suppose each matrix represents the augmented matrix for a system of linear equations. In each case, determine if the system is consistent. If the system is consistent, determine if the solution is unique.

a. $\begin{bmatrix} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 \end{bmatrix}$

b. $\begin{bmatrix} 0 & \blacksquare & * & * & * \\ 0 & 0 & \blacksquare & * & * \\ 0 & 0 & 0 & \blacksquare & 0 \end{bmatrix}$

Answer:

The matrix a. is consistent, since the last column is not a pivot column. However, it has infinitely many solutions since the third column is not a pivot column.

Problem 8 [8 pts] continued on next page...

The matrix b. is consistent, since the last column is not a pivot column. However, it has infinitely many solutions since the first column is not a pivot column.

Problem 9 [6 pts]

Mark each statement True or False. You do not need to justify each answer.

a. In some cases, a matrix may be row reduced to more than one matrix in reduced echelon form, using different sequences of row operations.

b. A basic variable in a linear system is a variable that corresponds to a pivot column in the coefficient matrix.

c. If one row in an echelon form of an augmented matrix is $\begin{bmatrix}0 & 0 & 0 & 5 & 0\end{bmatrix}$, then the associated linear system is inconsistent.

Answer:

The answers to these problems are, in order, F, T, F.

Problem 10 [10 pts]

Mark each statement True or False. You do not need to justify each answer.

a. The reduced echelon form of a matrix is unique.

b. If every column of an augmented matrix contains a pivot, then the corresponding system is consistent.

c. The pivot positions in a matrix depend on whether row interchanges are used in the row reduction process.

d. A general solution of a system is an explicit description of all solutions of the system.

e. Whenever a system has free variables, the solution set contains many solutions.

Answer:

The answers to these problems are, in order, T, F, F, T, F.

Problem 11 [4 pts]

Compute $\boldsymbol{u} + \boldsymbol{v}$ and $\boldsymbol{u} - 2\boldsymbol{v}$.

$$\boldsymbol{u} = \begin{bmatrix}-1\\2\end{bmatrix}, \quad \boldsymbol{v} = \begin{bmatrix}-3\\-1\end{bmatrix}$$

Answer:

$$\boldsymbol{u}+\boldsymbol{v}=\begin{bmatrix}-4\\1\end{bmatrix},\quad \boldsymbol{u}-2\boldsymbol{v}=\begin{bmatrix}5\\4\end{bmatrix}.$$

Problem 12 [4 pts]

Write a system of equations that is equivalent to the given vector equation.

$$x_1\begin{bmatrix}3\\-2\end{bmatrix}+x_2\begin{bmatrix}7\\3\end{bmatrix}+x_3\begin{bmatrix}-2\\1\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$$

Answer:

$$\begin{aligned}3x_1+7x_2-2x_3&=0\\-2x_1+3x_2+x_3&=0\end{aligned}$$

Problem 13 [4 pts]

Write a vector equation that is equivalent to the given system of equations.

$$\begin{aligned}x_2+5x_3&=0\\4x_1+6x_2-x_3&=0\\-x_1+3x_2-8x_3&=0\end{aligned}$$

Answer:

$$x_1\begin{bmatrix}0\\4\\-1\end{bmatrix}+x_2\begin{bmatrix}1\\6\\3\end{bmatrix}+x_3\begin{bmatrix}5\\-1\\-8\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}.$$

Problem 14 [8 pts]

Determine if $\boldsymbol{b}$ is a linear combination of the vectors formed from the columns of the matrix A.

$$A=\begin{bmatrix}1&-4&2\\0&3&5\\-2&8&-4\end{bmatrix},\quad \boldsymbol{b}=\begin{bmatrix}3\\-7\\-3\end{bmatrix}$$

Answer:

Problem 14 [8 pts] continued on next page...

We write down the augmented matrix:

$$\begin{bmatrix} 1 & -4 & 2 & 3 \\ 0 & 3 & 5 & -7 \\ -2 & 8 & -4 & -3 \end{bmatrix}$$

Reducing the matrix by the same method we used in previous problems, we obtain the reduced echelon form:

$$\begin{bmatrix} 1 & 0 & \frac{26}{3} & 0 \\ 0 & 1 & \frac{5}{3} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

which has a pivot column as the last column, and implies that $\boldsymbol{b}$ is not a linear combination of the column vectors of A.

Problem 15 [10 pts]

Let $\boldsymbol{a}_1 = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$, $\boldsymbol{a}_2 = \begin{bmatrix} -5 \\ -8 \\ 2 \end{bmatrix}$, and $\boldsymbol{b} = \begin{bmatrix} 3 \\ -5 \\ h \end{bmatrix}$. For what value(s) of h is $\boldsymbol{b}$ in the plane spanned by $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$?

Answer:

Assume $\boldsymbol{b} = x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2$. We obtain the equations:

$$\begin{aligned} x_1 - 5x_2 &= 3 \\ 3x_1 - 8x_2 &= -5 \end{aligned}$$

Solving the equations by the same method we used in previous problems, we obtain the solution:

$$\begin{cases} x_1 = -7, \\ x_2 = -2. \end{cases}$$

Therefore, we obtain $h = -x_1 + 2x_2 = 3$.

Problem 16 [8 pts]

Let $\boldsymbol{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $\boldsymbol{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Show that $\begin{bmatrix} h \\ k \end{bmatrix}$ is in $\mathrm{Span}\{\boldsymbol{u}, \boldsymbol{v}\}$ for all h and k.

Answer:

We write down the augmented matrix:

$$\begin{bmatrix} 2 & 2 & h \\ -1 & 1 & k \end{bmatrix}$$

Reducing the matrix by the same method we used in previous problems, we obtain the reduced echelon form:

$$\begin{bmatrix} 1 & 0 & \frac{1}{4}h - \frac{1}{2}k \\ 0 & 1 & \frac{1}{4}h + \frac{1}{2}k \end{bmatrix}.$$

Therefore, we have

$$\left(\frac{1}{4}h - \frac{1}{2}k\right)\boldsymbol{u} + \left(\frac{1}{4}h + \frac{1}{2}k\right)\boldsymbol{v} = \begin{bmatrix} h \\ k \end{bmatrix},$$

which implies $\begin{bmatrix} h \\ k \end{bmatrix}$ is in $\text{Span}\{\boldsymbol{u}, \boldsymbol{v}\}$.

Problem 17 [4 pts]

Use the vector $\boldsymbol{u} = (u_1, \ldots, u_n)$ to verify the following algebraic properties of $\mathbb{R}^n$.

a. $\boldsymbol{u} + (-\boldsymbol{u}) = (-\boldsymbol{u}) + \boldsymbol{u} = \boldsymbol{0}$

b. $c(d\boldsymbol{u}) = (cd)\boldsymbol{u}$ for all scalars c and d

Answer:

For property a., we have $\boldsymbol{u}+(-\boldsymbol{u}) = (u_1-u_1, \ldots, u_n-u_n) = \boldsymbol{0}$, and $(-\boldsymbol{u})+\boldsymbol{u} = (-u_1+u_1, \ldots, -u_n+u_n) = \boldsymbol{0}$.

For property b., we have $c(d\boldsymbol{u}) = (c \cdot du_1, \ldots, c \cdot du_n)$, and $(cd)\boldsymbol{u} = (cd \cdot u_1, \ldots, cd \cdot u_n)$, which implies $c(d\boldsymbol{u}) = (cd)\boldsymbol{u}$.

Problem 18 [8 pts]

Compute the following products using (a) the definition, as in Example 1, on page 35 of the book *Linear Algebra and its applications, Forth edition by David C. Lay*, and (b) the row-vector rule for computing $A\boldsymbol{x}$, as in Example 5, on page 38 of the book *Linear Algebra and its applications, Forth edition by David C. Lay.* If a product is undefined, explain why.

1. $\begin{bmatrix} -4 & 2 \\ 1 & 6 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 7 \end{bmatrix}$

2. $\begin{bmatrix} 1 & 3 & -4 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

Answer:

Multiplication 1. is undefined since the matrix has 2 columns while the vector has 3 rows.

For multiplication 2., the two ways can be written as:

$$1 \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 2 \cdot \begin{bmatrix} 3 \\ 2 \end{bmatrix} + 1 \cdot \begin{bmatrix} -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 8 \end{bmatrix},$$

and

$$\begin{bmatrix} 1 \cdot 1 + 3 \cdot 2 - 4 \cdot 1 \\ 3 \cdot 1 + 2 \cdot 2 + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 8 \end{bmatrix}.$$

Problem 19 [4 pts]

Use the definition of $A\boldsymbol{x}$ to write the following vector equation as a matrix equation.

$$x_1 \begin{bmatrix} 4 \\ -1 \\ 7 \\ -4 \end{bmatrix} + x_2 \begin{bmatrix} -5 \\ 3 \\ -5 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 7 \\ -8 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ -8 \\ 0 \\ -7 \end{bmatrix}$$

Answer:

$$\begin{bmatrix} 4 & -5 & 7 \\ -1 & 3 & -8 \\ 7 & -5 & 0 \\ -4 & 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -8 \\ 0 \\ -7 \end{bmatrix}.$$

Problem 20 [4 pts]

Write the following system first as a vector equation and then as a matrix equation.

$$\begin{aligned} 5x_1 \; &+x_2 \; -3x_3 = 8 \\ &+2x_2 \; +4x_3 = 0 \end{aligned}$$

Answer:

$$x_1 \begin{bmatrix} 5 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \end{bmatrix},$$

$$\begin{bmatrix} 5 & 1 & -3 \\ 0 & 2 & 4 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \end{bmatrix}.$$

Problem 21 [10 pts]

Let $\boldsymbol{v}_1 = \begin{bmatrix} 0 \\ 0 \\ -3 \end{bmatrix}$, $\boldsymbol{v}_2 = \begin{bmatrix} 0 \\ -3 \\ 9 \end{bmatrix}$, $\boldsymbol{v}_3 = \begin{bmatrix} 4 \\ -2 \\ -6 \end{bmatrix}$. Does $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$ span $\mathbb{R}^3$? Why or why not?

Answer:

For an arbitrary vector $\begin{bmatrix} x & y & z \end{bmatrix}^\top \in \mathbb{R}^3$, we try spanning it by $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$. We obtain

$$\begin{bmatrix} 0 & 0 & 4 & x \\ 0 & -3 & -2 & y \\ -3 & 9 & -6 & z \end{bmatrix},$$

Problem 21 [10 pts] continued on next page…

and its reduced echelon form is

$$\begin{bmatrix} 1 & 0 & 0 & -x-y-\frac{1}{3}z \\ 0 & 1 & 0 & -\frac{1}{6}x-\frac{1}{3}y \\ 0 & 0 & 1 & \frac{1}{4}x \end{bmatrix}.$$

Therefore, $\begin{bmatrix} x & y & z \end{bmatrix}^\top$ is in $\text{Span}\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$, which implies $\text{Span}\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\} = \mathbb{R}^3$.

Problem 22 [10 pts]

Mark each statement True or False. Justify each answer.

a. The equation $A\boldsymbol{x} = \boldsymbol{b}$ is referred to as a *vector equation*.

b. A vector $\boldsymbol{b}$ is a linear combination of the columns of a matrix A if and only if the equation $A\boldsymbol{x} = \boldsymbol{b}$ has at least one solution.

c. The equation $A\boldsymbol{x} = \boldsymbol{b}$ is consistent if the augmented matrix $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}$ has a pivot position in every row.

d. If the columns of an $m \times n$ matrix A span $\mathbb{R}^m$, then the equation $A\boldsymbol{x} = \boldsymbol{b}$ is consistent for each $\boldsymbol{b}$ in $\mathbb{R}^m$.

e. If A is an $m \times n$ matrix and if the equation $A\boldsymbol{x} = \boldsymbol{b}$ is inconsistent for some $\boldsymbol{b}$ in $\mathbb{R}^m$, then A cannot have a pivot position in every row.

Answer:

For statement a., the answer is false, since the equation is referred to as a matrix equation.

For statement b., the answer is true, since any solution to $A\boldsymbol{x} = \boldsymbol{b}$ matches a valid choice of the coefficients of the linear combination.

For statement c., the answer is false, since consistency requires the last column not to be a pivot column.

For statement d., the answer is true; the reason is the same as in statement b.

For statement e., the answer is true, since if A had a pivot position in every row, then $A\boldsymbol{x} = \boldsymbol{b}$ would be consistent for every $\boldsymbol{b}$.